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ORIGINAL RESEARCH

Robust 3D brain segmentation in magnetic resonance image with weighted feature fusion

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Abstract

Accurate tissue segmentation in brain magnetic resonance image is of paramount importance for brain reconstruction and analysis. Supervoxels analysis enables relatively robust segmentation of brain tissues, by assigning the supervoxels to tissue clusters according to the inherent information of multiple features captured from the supervoxels. In this popular framework, difficulties are encountered in clustering imposed by the insufficient selection, deficient utilisation of features, and rough supervoxel boundaries due to the influence of noise. To address these aforementioned challenges, the authors propose a weighted feature fusion clustering segmentation method for dividing supervoxels into three tissues on the basis of a robust feature fused similarity matrix. To comprehensively capture the similarities of features among the supervoxels, we construct three complementary similarity matrices on the basis of multiple features from three aspects, that is, appearance, shape, and space location. Next, the authors propose a weighted similarity network fusion method to discriminatively fuse the three similarity matrices into one single matrix. This novel fusion process cannot only extract the common and complementary information of features, but also automatically adjust the weight of similarities according to its reliability. Finally, we achieve initial segmentation results by performing spectral clustering on the fused similarity matrix. To optimise the segmented boundaries affected by noise in the initial segmentation results, the authors construct a voxel-wise diffusion energy function by considering the intensity information of neighbouring voxels. Experiments on two public brain magnetic resonance image datasets demonstrate the robustness of the proposed approach.

1 | INTRODUCTION

Brain MR image tissue segmentation is performed to divide the brain structure into grey matter (GM), white matter (WM), and cerebrospinal fluid (CSF) from the magnetic resonance (MR) images. Segmentation is a key step for the construction, analysis, and understanding of the brain [1–3]. Traditional brain MR image is mainly manually segmented by experts and thus is time-consuming and cannot effectively deal with a large number of brain MR images. At present, automatic segmentation in brain MR images has been proposed. Parametric methods [4, 5], non-parametric approaches [6, 7], and atlas-based strategies [8] have been developed gradually for this segmentation task. Usually the images are 2-dimensional (2D), where the MRI images are usually presented in slices from top to bottom. Earlier approaches focus on pixel-wise segmentation. Pixel-wise segmentation methods are performed on a single slice of an MR image [9], and all segmented slices are combined by using the connective information. Recent years, by using sophisticated computer calculation, the 2D brain image slices can be joined together to produce a 3-dimensional (3D) image. Given that above methods lack the utilisation of 3D neighbouring information, it is difficult to guarantee the continuity of tissues between adjacent slices, although they could achieve relatively ideal segmentation result on one single slice. Voxel-wise techniques can solve the problem of achieving discontinuous tissues in pixel-wise segmentation methods [10–12]. However, voxel-wise segmentation neglects spatial information and does not efficiently use local image features. Moreover, given the large amounts of voxels involved in 3D MR image, these methods always require a high computational cost.

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Recent years, several techniques using deep-convolution networks have also been proposed in the past for brain MR image segmentation [13]. However, before the actual process for segmentation task, these methods require training through which its free parameters are updated and learned via known examples; resulting relatively larger time complexity.

'Supervoxel' is a set of superpixels with similar intensity in the 3D volume. Supervoxel techniques not only provide a convenient use of the local features in a 3D image, but also reduce the computational cost of the subsequent processing tasks. Therefore, supervoxel-based analysis has become a prevalent preprocessing method in medical image analysis [14], in which voxels with similar features are assigned into a meaningful supervoxel by maintaining region homogeneity. Superpixels generation methods could be classified as graph-based and clustering-based [15, 16]. Specifically, graph-based methods consider each pixel as a node in a graph, and the edge weight is the similarity between neighbours. By minimising a cost function, the superpixels are generated, such as Topology Preserved Regular Superpixel (TPS) [17], Lazy Random Walks Superpixel (LRW) [18] and so on. However, the computational costs of this kind methods are high and thus inadequate for large-size images especially for 3D images. Clustering-based methods divide pixels into different clusters and refine the boundaries of them iteratively, until satisfying the convergence criteria. Due to most clustering-based methods performing directly on pixels in image, the time costs of them are less than graph-based methods [15]. However, some of them can only generate regular superpixels and cannot adhere to the boundaries well, such as Compact Watershed (CW) [19], Contour Relaxed Superpixels (CRS) [20], Voxel Cloud Connectivity Segmentation-supervoxels (VCCS) [21], Iterative Spatial Fuzzy Clustering Supervoxel (ISFCS) [14], which are not suitable for generating supervoxels in the brain volume due to its complexity boundaries. Among these supervoxel generation methods, simple linear iterative clustering (SLIC) [22] is a highly efficient algorithm that can generate supervoxels with compactness, by using k-means clustering to group neighbouring voxels into seeds sampled uniformly. Furthermore, SLIC is widely used in pre-processing medical images due to its good adhesion of boundary in complex images.

Although the supervoxels can utilise the local features and spatial information, certain issues still need to be noted. First, comprehensive and useful selections of the informative features are important to improve the segmentation accuracy. The features are expected not only to reflect the general characteristics but also contain boundary detail information of the supervoxels, to describe a supervoxel from well-rounded aspects. Second, the well-rounded features likely include several common and complementary characteristics of supervoxels. Fully utilising and capturing the common and complementary characteristics of all features are also an important step. Third, certain boundaries of supervoxels might be influenced by noise in brain MR images [23, 24], causing rough boundaries after clustering. Therefore, accurately optimising these noisy boundaries is crucial to improve the robustness of tissue segmentation.

To deal with these issues, we propose a weighted feature fusion clustering segmentation method, which is a robust clustering method based on comprehensive feature capturing and utilising. Specifically, after the prevalent supervoxel-level preprocessing, we extract the features of each supervoxel from three aspects, that is, appearance, shape, and space location, which could reflect both general and detailed information of supervoxels. To measure the similarity of each paired supervoxels, we construct three similarity matrices based on these three complementary features, by using different appropriate similarity measurements. Specifically, similarities with regard to appearance, shape, and spatial features are measured by Spearman's correlation, Euclidean distance, and k-means clustering, respectively. To fully and robustly capture common and complementary characteristics of these three similarity matrices, we propose a weighted similarity network fusion method to fuse such matrices into one single similarity matrix. During the fusion process, we design a weight to automatically adjust feature similarities of each paired supervoxels according to the noise influencing, by considering the supervoxel-level intensity inhomogeneity factor. Compared with other state-of-the-arts segmentation methods performed on two public brain MR image datasets, our novel algorithm is noise tolerant and can generate more accurate segmentation results.

In general, our contributions are listed as follows:

(1) To completely capture the features of supervoxels, for each two-paired supervoxels, we propose to construct its similarity matrix by using appropriate similarity measurements on the basis of its characteristics. The similarity matrices are defined based on their appearance, shape, and spatial features, which not only contain the general intensity information but also include the boundary details.
(2) To fully utilise and capture the common and complementary characteristics of all features, we propose a new robust weighted similarity network fusion method to discriminatively fuse the similarity matrices based on these three features into one single matrix.
(3) To overcome the issue that certain boundaries of supervoxels might be influenced by noise, we propose a new diffusion method to post-process the segmentation results by constructing a diffusion energy function. The energy function combines both the voxel-wise and supervoxel-wise neighbouring information, to optimise the initial segmentation results.

2 | METHODS

We propose a supervoxel-based weighted feature fusion clustering segmentation method (SWFFS) for noised brain MR image. Firstly, we generate supervoxels by using a classic and popular SLIC method. Then, for each supervoxel, we extract its features of appearance, shape and space location. Based on the supervoxel-level features, we construct the similarity matrices to measure the similarity of each paired supervoxels. Next, we

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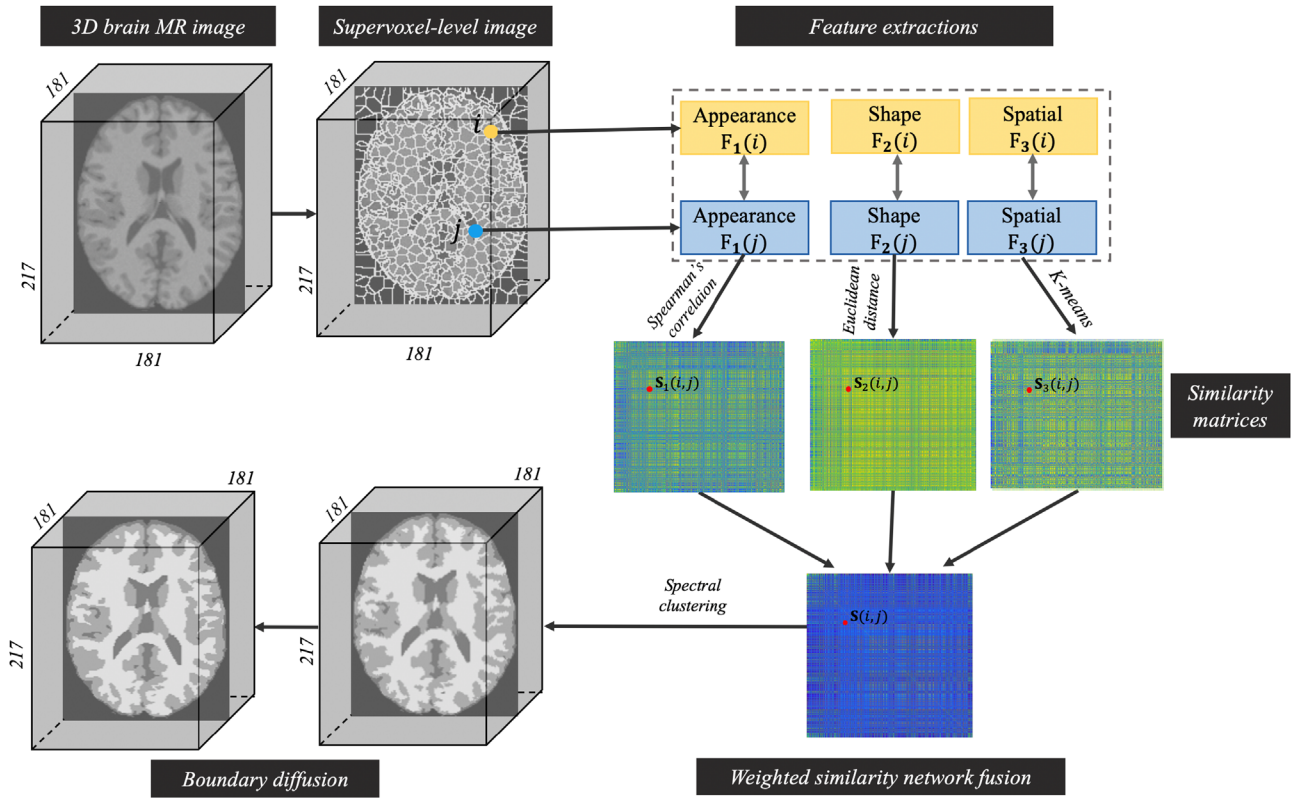


FIGURE 1 The framework of our segmentation method

fused these similarity matrices and run the spectral clustering on the fused similarity matrix. Finally, we construct an energy function to diffuse the rough boundaries in the initial segmentation results, by considering the neighbouring voxels information. The pipeline of SWFFS is shown in Figure 1. We describe each step in detail in this section.

2.1 | Extracting supervoxel-level features

Given that our segmentation method is processed on the basis of supervoxel level, we first generate the 3D supervoxels in brain MR images, by adopting the SLIC algorithm, which is a modification of k-means clustering approach. One parameter should be specified in SLIC, that is, the number of supervoxels. SLIC iteratively updates the previous centres to the best matching voxels on the basis of similarities of intensity and spatial information, thereby simultaneously ensuring compactness and homogeneity. Different from SLIC applied on 2D superpixel generation, handling 3D supervoxels by SLIC is needed a consideration of the depth dimension to the spatial similarity term. For example, the size of 3D brain MR image from BrainWeb dataset is 181 × 217 × 181. Here, the first 181 is the number of slices, 217 and the last 181 is the height and width of each slice, respectively. In this case, the number of slices (depth information) should also be considered into measuring the spatial similarity. At each iteration, voxels are assigned to the supervoxel with the nearest centre. When the difference between the new centres and the previous ones is smaller than a threshold, the iteration stops. Note that, considering reducing the time cost, the supervoxels are generated only in the brain region in our methods, as the operation in previous study [25]. Specifically, we only remain the brain volume part by clipping the cuboid whose voxels are all background. As shown in Figure 2, the supervoxels can adhere well to the boundaries among different tissues in axial, sagittal, and coronal views.

Subsequently, each supervoxel is assigned to its appropriate tissue category according to the similarity of its features; hence, this model is considered as a clustering model. As more useful features are utilised in the clustering, the tissue segmentation results will be more precise. General and detailed information of supervoxels is expected to be captured. To comprehensively extract the well-rounded characteristics of each supervoxel, we

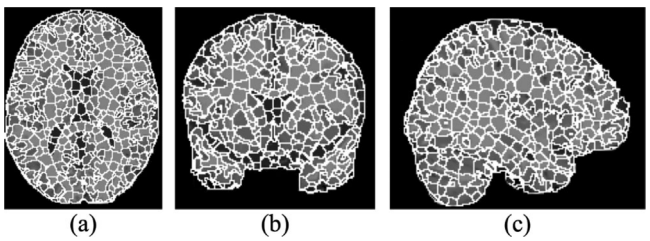


FIGURE 2 An example of supervoxels generated by the SLIC algorithms. (a) Axial view, (b) sagittal view, and (c) coronal view

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TABLE 1 List of features used in this work

Type	Feature	Dimension
Appearance	histogram of intensity within the supervoxel	50
Shape	3D SIFT key voxels of supervoxel	10
Space location	x; y; z coordinate difference from the supervoxel centre to the image center	3

propose to define features from three aspects, that is, appearance, shape, and space location, as shown in Table 1.

Specifically, to describe the appearance of each supervoxel, we compute the intensity histogram $F_1(i)$ for the supervoxel i, and the number of bins in histogram is set as 50. Theoretically, the number of histogram bins is larger, the features can compress more detail information of MR image. However, finer bins are always along with the increasing of sensitivity to noise. The ideal set should trade off the detail preservation and noise influence. To find the best set, we have tried different numbers of bins in [10, 70], and find 50 is the most suitable set with best segmentation performance. The general characteristics of this supervoxel can be described because $F_1(i)$ includes the intensity information of all voxels in the supervoxel i.

To describe the shape feature for the supervoxel i, we construct a vector $F_2(i)$ consisting of the intensities of the key voxels selected by 3D scale invariant feature transform (SIFT) [26]. The 3D SIFT method detects the outstanding and stable key voxels by finding the scale-space extreme values of the difference-of-Gaussian (DoG), with high enough contrast within the neighbourhood of the current key voxels. These key voxels selected by 3D SIFT are the salient voxels in terms of the intensity and spatial locations, such as the voxels near the boundaries among different tissues. To certain extent, the shape feature can reflect the boundary detail information in each supervoxel. Therefore, the appearance and shape features of supervoxels are complementary. The appearance feature can reflect the general intensity distribution of supervoxel, whereas the shape feature can capture the boundary details of supervoxel. The dimension of shape is set according to the minimum number of key-voxels that can be detected by 3D SIFT in all supervoxels, which is 10 in our case.

To measure the 3D dimensional position of supervoxel i, we define the spatial feature $F_3(i)$ as the coordinate difference between the average coordinate of all voxels in the supervoxel i and the coordinate of central voxel. The central voxel is the centre of whole brain MR image.

2.2 | Constructing similarity matrices of features

The similarities of supervoxels can be measured using the similarities of these features between paired supervoxels. In order to capture accurate feature similarities among different supervoxels, we choose appropriate similarity measurements for different

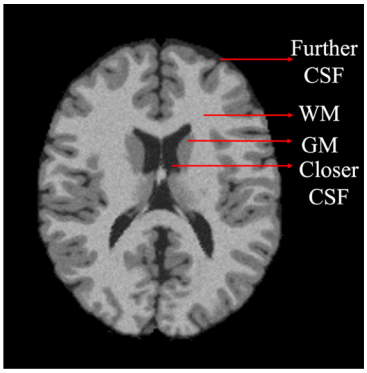


FIGURE 3 An example of different brain tissues

features. The appearance feature F_1 is defined by using the histograms, which shows the intensity distribution of all voxels in each supervoxel. When two histograms are similar, their distributions are supposed to be proportional or monotonic. Spearman's correlation coefficient is an effective measurement of a monotonic relationship, and it is sensitive to the order of elements. Therefore, using Spearman's correlation is suitable to measure the similarity between intensity histograms of different supervoxels. Between each pair of supervoxels i and j, we compute the Spearman's correlation coefficient p of their appearance features and achieve their similarity as follows:

$$S_1(i,j) = \frac{1 + p(F_1(i), F_1(j)}{2}, i, j \in 1, .., N. \quad (1)$$

Here, N is the total number of supervoxels in brain MR image. S_1 ranges from 0 to 1. Intuitively, high correlation between two supervoxels indicates high similarity of their intensity histograms, and similar intensity distribution.

The shape feature F_2 is defined by using the key voxels of supervoxels. The similarity between key points of different supervoxels is independent from the order of key points. Therefore, we adopt the simple Euclidean distance to measure the similarity of shape features between each paired supervoxels. Specifically, between each pair of supervoxels i and j, the Euclidean distance of their shape features $F_2(i)$ and $F_2(j)$ is denoted as $\mathrm{dist}(i,j)$, and the similarity S_2 is defined as

$$S_2(i,j) = 1 - \frac{\mathrm{dist}(i,j)}{\mathrm{dist}_{\max}}, i, j \in 1, .., N. \quad (2)$$

Here, $\mathrm{dist}_{\max}$ is the maximum dist among all supervoxels. S_2 ranges from 0 to 1. High value between two supervoxels indicates high similarity of their intensities of key voxels and similar boundary detail information.

The space location feature F_3 indicates the 3D dimensional position relative to the central voxel. In this part, we perform k-means clustering on the space location feature F_3, to get the similarity between each pair supervoxels. The brain structures without skull are the CSF, WM, GM, and further CSF from the nearest to the furthest regions from central voxel in the whole MR image, as shown in Figure 3. Therefore, the number of

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clusters is set as 4, to divide the closer CSF, WM, GM and further CSF in brain region. The space location feature of each supervoxel is updated to the centre of cluster that this supervoxel belonging to in k-means results. Note that the space location features of supervoxels in further CSF region are updated to those of the supervoxels in the closer CSF region. Between each pair of supervoxels i and j, the Euclidean distance of their updated space location features $F_3(i)$ and $F_3(j)$ is simply represented as dist2(i, j). Through this, when supervoxels i and j are belonged into the same cluster, the distance of them dist2(i, j) is 0. Note that the background is not included in the clustering. Finally, the similarity between $F_3(i)$ and $F_3(j)$ is defined as

$$S_3(i,j) = 1 - \frac{\text{dist2}(i,j)}{\text{dist2}_{\max}}, i, j \in 1, .., N. \tag{3}$$

Here, $\text{dist2}_{\max}$ is the maximum dist2 among all supervoxels. S_3 ranges from 0 to 1. High value between two supervoxels indicates close distance relative to the central voxel.

2.3 | Robust weighted similarity network fusion

A straightforward approach to divide the supervoxels into different tissue clusters according to complementary similarity matrices S_1, S_2 and S_3 is to perform clustering on the averaged matrix. However, simple averaging cannot fully capture complementary and common information of these three similarity matrices, and might lead to inaccurate segmentations. Zhu *et al.* [27] proposed a novel multi-modality medical image fusion method based on phase congruency and local Laplacian energy. Although it can achieve competitive performance for the fusion of whole multi-modality medical images, it cannot be applied directly in the fusion of our similarity matrices. To obtain well-rounded information from the similarity matrices, Wang *et al.* proposed the similarity network fusion method (SNF) [28]. However, directly using SNF in our work is not suitable, because it might also undesirably capture the information of noise. To address this issue, we propose a robust weighted similarity network fusion method to distinctively fuse these three similarity matrices into one single matrix S. As we introduced, the shape feature is achieved by using SIFT 3D method, which captures the salient information of supervoxels, such as the boundary or outstanding shape patterns, by using the key voxels in terms of intensity. To some extent, the shape feature is more sensitive to noises. When the supervoxel is influenced by noises, the shape feature may be unreliable. Therefore, we design a weight function to adaptively set to adjust the similarities of shape feature by considering the influence of noise in each supervoxel. Specifically, for each S_2, we first compute a full kernel matrix P_2 as follows:

$$P_2(i,j) = \begin{cases} \dfrac{w(i,j)\cdot S_2(i,j)}{2\sum_{k\neq i} w(i,k)\cdot S_2(i,k)}, & j \neq i \\ \dfrac{1}{2}, & j = i \end{cases}; \tag{4}$$

$$w(i,j) = \begin{cases} 1 + \dfrac{\delta_i \times \delta_j}{2(\delta_i + \delta_j)}, & if\ \ \sigma_i \leq \bar{\sigma}\ \ and\ \ \sigma_j \leq \bar{\sigma} \\ 1 - \dfrac{\delta_i \times \delta_j}{2(\delta_i + \delta_j)}, & if\ \ \sigma_i > \bar{\sigma}\ \ or\ \ \sigma_j > \bar{\sigma} \end{cases}, j \neq i; \tag{5}$$

$$\delta_i = \frac{\sigma_i - \sigma_{\min}}{\sigma_{\max} - \sigma_{\min}}. \tag{6}$$

Here, σ_i is the standard deviation of the intensity of voxels in supervoxel i. $\sigma_{\min}$, $\sigma_{\max}$ and $\bar{\sigma}$ are the minimum, maximum and average standard deviations of all supervoxels, respectively. w is a robust weight computed on the basis of the standard deviation of supervoxels i and j, ranging from 0.75 to 1.25. Note that the weights in the fusion process is not set manually, but adaptively. The weight is computed for each supervoxel i by utilising the standard deviation of the intensity of voxels in supervoxel i. It remains the same value during different iterations, because the voxel value is not changed during iteration. When one of the standard deviations of supervoxels i or j is higher than the average, this supervoxel might be influenced by more noises or intensity inhomogeneities. In this case, the similarity value $S_2(i, j)$ might be influenced by these issues and is thus unreliable; therefore, we decrease the influence of this similarity by multiplying the robust weight $w(i, j)$ lower than 1. At this time, the similarity of appearance and spatial features will help to estimate the influence of noise on this supervoxel, and the final fused similarity is mainly led by the other two full kernel matrices of appearance and spatial features. By contrast, when the standard deviations of supervoxels i and j are both lower than the average, the intensities are more homogenous in these supervoxels. In this case, the similarity value might be more reliable; therefore, we increase the influence of this similarity by multiplying a robust weight $w(i, j)$ larger than 1. P_2 is a full kernel normalised weight matrix. The normaliSation not only ensures $\sum_j P_m(i, j) = 1$, but also guarantees that P_2 avoids the self-similarity scale in the diagonal entries. Considering that S_1 and S_3 are relatively independent from the influence of noise, P_1 and P_3 are defined as follows:

$$P_m(i,j) = \begin{cases} \dfrac{S_m(i,j)}{2\sum_{k\neq i} S_m(i,k)}, & j \neq i \\ \dfrac{1}{2}, & j = i \end{cases}; m \in \{1, 3\}. \tag{7}$$

Next, we calculate a sparse kernel matrix Q_m as follows:

$$Q_m(i,j) = \begin{cases} \dfrac{S_m(i,j)}{2\sum_{k\neq N_i} S_m(i,k)}, & j \neq N_i \\ 0, & otherwise \end{cases}; m \in \{1, 2, 3\}; \tag{8}$$

Here, N_i represents the K most similar supervoxels for supervoxel i, calling the neighbourhood of supervoxel i. Compared with the non-neighbouring supervoxels, the neighbouring supervoxels with high similarities are more reliable. In this

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present work, we set the similarity between non-neighbouring supervoxels as zero in the sparse kernel matrix Q_m. Therefore, P_1, P_2 and P_3 include full information of the similarities of each supervoxel to all other supervoxels on the basis of features F_1, F_2 and F_3, whereas Q_1, Q_2 and Q_3 only include the similarities to the K most similar supervoxels for each supervoxel. To fuse the three similarity matrices, for iteration t, $P_m^t, m \in \{1, 2, 3\}$ is updated as

$$P_m^t = Q_m \times \left(\frac{\sum_{k \neq m} P_k^{t-1}}{2} \right) \times (Q_m)^T, m \in \{1, 2, 3\}. \quad (9)$$

Here, $(\cdot)^T$ represents matrix transpose. In Equations (8) and (9), the three matrices P_1, P_2 and P_3 are fused via a message-passing process wherein these three matrices are modified iteratively to become more similar in each iteration. In this process, low similarities are gradually eliminated, and high similarities appear in at least one matrix are added to others. The final fused matrix S is the average of $P_1^{(t^*)}$, $P_2^{(t^*)}$ and $P_3^{(t^*)}$ after t^* iterations.

During the fusion process, we design a weight to measure the intensity inhomogeneity of the supervoxels that can automatically decrease/increase the influence of the unreliable/reliable similarities of the shape feature in fusion. If two supervoxels are really similar but affected by noise, their similarity in P_2 might be undesirably decreased. In this case, P_1 can help maintain their high similarity during the fusion process, and vice versa. Moreover, because Q_m is defined by using K most similar neighbours of supervoxel i in P_m (in other words, Q_m is a graph of P_m), which can reduce some noise between instances. Therefore, the weighted SNF is more robust to the noise in the fusion process. Note that only two supervoxels influenced by noise have very similar appearance, spatial and shape features, their similarity would be high. The possibility of this case is very low because there are many voxels in one supervoxel, and the influence of noise is random.

2.4 | Clustering and boundary diffusion

Given the fused similarity matrix S, we perform the spectral clustering on it to achieve the initial segmentation results. To better capture the input data point distributions, spectral clustering only utilises the top eigenvectors in the eigenspace of the similarity matrix. As spectral clustering requires a pre-defined number of clusters, we set the number of clusters as 3.

Considering that the SLIC method defines the boundary by using the intensity, some boundaries are influenced by the noises in the initial segmentation results. Combining the effect of neighbourhood will help to decrease the effect of noise and correct the segmentation results based on supervoxels. In the present study, we propose a post-processing to diffuse the initial segmentation boundary that not only uses supervoxel-level information but also combines voxel-level neighbouring intensity information. Specifically, we construct a diffusion energy function that utilises the neighbourhood information for each voxel to decrease the noise or intensity inhomogeneous influence. The neighbouring information is iteratively combined by minimising the energy function to diffuse the boundaries in the initial segmentation results. The energy function is defined as follows:

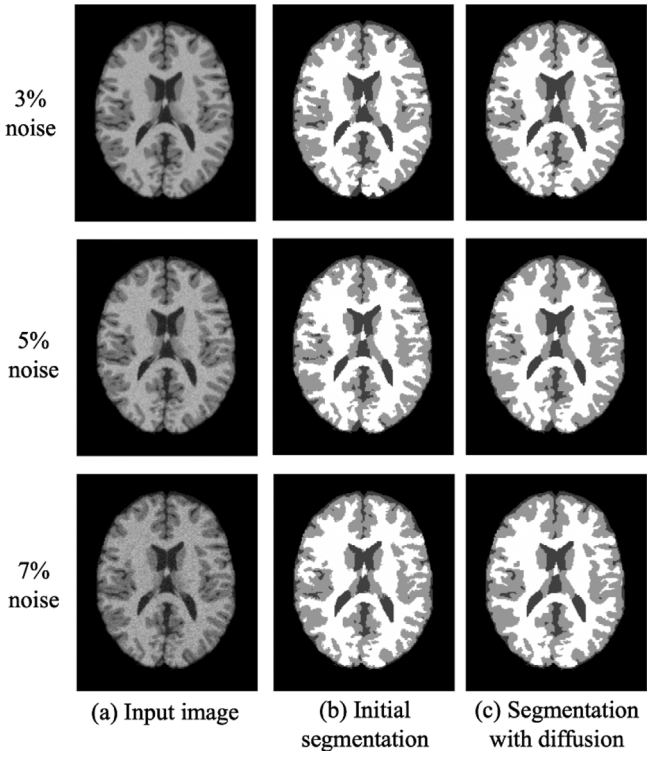


FIGURE 4 Boundary diffusion performed on images influenced by different levels of noises. (a) Input image, (b) initial segmentation results, and (c) optimised segmentation results

$$E = \sum_{i \in M} \|v_i - c(i)\| + \sum_{i,j \in R_i} \alpha \|c(i) - c(j)\|. \quad (10)$$

Here, M is the number of voxels in brain MR image. v_i is the intensity of the voxel i, and R_i is the spatial neighbourhoods of the voxel i. $c(i)$ is the centre intensity of the cluster that voxel i assigned to. The center intensity of each cluster is the average intensity of all voxels belonging to this cluster. σ is a smooth parameter to control the weight of neighbouring information. Specifically, the first term in Equation (10) is a data term to adaptively keep v_i close to $c(i)$, and the second term is a smoothness term to eliminate the influence of the intensities of unreliable and noisy voxels by propagating the reliable intensities from neighbours. It can propagate the reliable cluster information from neighbouring voxels to reduce the influence of noise. The energy function in Equation (10) is solved by the graph cuts method to obtain the final segmentation result. After this post-processing, some rough boundaries that influenced by noises are smoothed, as shown in Figure 4.

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3 | EXPERIMENTS

In the present study, experiments of the proposed segmentation algorithm are performed on two publicly available benchmark datasets. The first one is BrainWeb [29], which consists of six simulated brain MR volumes influenced by different levels of noise (https://brainweb.bic.mni.mcgill.ca), with a size of 181 × 217 × 181 voxels. The second one is IBSR-18, which contains 18 real brain MR volumes and is available at Internet Brain Segmentation Repository (www.cma.mgh.harvard.edu/ibsr/), with a size of 256 × 256 × 128 voxels. All images are provided with their ground truth segmentations of WM, GM and CSF tissues. According to previous research on supervoxels applied on brain image segmentation [14], the number of supervoxels is empirically set to 5000. Moreover, according to our other research on diffusion method [30], the smooth parameter α is set as 0.5 empirically. The experiments were performed on a macbook pro with Intel Core 2.5 GHz and a 16 GB RAM, with NVIDIA GeForce GT 750M 2 GB.

The performance of the proposed SWFFS is compared with several state-of-the-arts algorithms, including fuzzy c-means (FCM) [31], FMRIB Software Library (FSL) version 6.0.3 [32], statistical parameter mapping (SPM) version 12 [33], rough fuzzy C-means with spatial constraint (RFCMSC) [4] and adaptive weighted spatial FCM (AWSFCM) [34] performed on the simulative and real brain MR images. FCM is a widely used medical image segmentation algorithm that allows one pixel to belong to multiple classes concurrently. FSL and SPM are two well-known and widely used brain MR volume analysis tools. RFCMSC and AWSFCM are two up-to-data segmentation methods for brain MR image presented in 2018 and 2020 years, respectively. The segmentation methods in both FMRIB Software Library (FSL) and statistical parameter mapping (SPM) version 12 toolboxes are mainly based on adult brain MR segmentation templates, and there is no parameter to be set by ourselves [32] [33]. In RFCMSC, there is a parameter to control the effect of neighbouring term, which is set as 0.8 empirically [4] . In AWSFCM, the parameter is adaptively set according to voxel values [34].

Figure 5 visualises the brain MR image segmentation results of FCM, FSL, SPM, RFCMSC and SWFFS. The three input images are all from BrainWeb, and influenced by 3%, 5%, and 7% noises. The grey value distribution of the noise in the simulation image follows a Gaussian density function, whose average value is 0. The standard deviation of this Gaussian density function is the product of the noise level and the average of the grey value of the WM. As shown in Figure 5, SWFFS can achieve more robust segmentation results compared with FCM, FSL, and SPM. To visualise our 3D result, we render the boundary between WM and GM in segmentation results to 3D brain surface, as shown in Figures 6 and 7. The surfaces are shown from sagittal and axial views, respectively. Specifically, when the level of noise increases, the segmentation results of FCM and FSL methods become more easily influenced by noise, whereas SPM, RFCMSC and SWFFS can obtain more robust segmentation results. When the images are influenced by extremely large noises, that is, 7% noise, some boundaries between WM and GM become unclear in the segmentation results of SPM and RFCMSC, as shown in Figures 6 and 7; at this time, the segmentation results of SWFFS are smoother, and the boundaries among tissues are clearer. Moreover, the boundaries between the WM and GM near the subcortical regions and the frontal cortex are closer to the ground truth, compared with those obtained by FSL, SPM and RFCMSC.

Figure 8 visualises the tissue segmentation results of the No. 8 volume in IBSR-18 dataset from the axial, sagittal, and coronal views. In all three views, the segmentation results of SWFFS are closer to the ground truth, particularly in the regions near the subcortex. The segmentation results of FCM and FSL are influenced by noise and the boundaries are rougher. The segmentation results of SPM and RFCMSC present smoother boundaries and more robust results. However, the segmentation results achieved by SPM and RFCMSC present thinner and thicker GM near the black background, respectively, especially in the sagittal and coronal views. The segmentation results of SWFFS can achieve more accurate tissue boundaries and more robust results.

We adopt the segmentation accuracy (SA) to assess the segmentation results quantitatively. SA is defined as the ratio between the number of voxels clustered correctly and the total number of voxels. Formally,

$$SA = \sum_{k=1}^{C} \frac{A_k \cap C_k}{\sum_{i=1}^{C} C_i}. \tag{11}$$

where A_k is the set of voxels belonging to the k class in the result, and C_k is the set of voxels belonging to the k class in the ground truth. We compute the total SA values and the SA values of each tissue segmented by performing the four algorithms on BrainWeb and IBSR datasets, and the results are shown in Tables 2 and 3, respectively. In Table 2, SWFFS can

TABLE 2 Comparative SA analysis of different algorithms on brainweb

Noise	Tissue	FCM	FSL	SPM	AWSFCM	RFCMSC	SWFFS
3%	WM	0.8126	0.7893	0.8876	0.9245	0.8903	0.9467
	GM	0.8377	0.7394	0.8992	0.8903	0.8857	0.9190
	CSF	0.9207	0.9305	0.9462	0.9483	0.9549	0.9680
	Total	0.9008	0.9031	0.9426	0.9511	0.9500	0.9671
5%	WM	0.8076	0.7533	0.8688	0.9235	0.8892	0.9371
	GM	0.8217	0.7356	0.8683	0.8864	0.8601	0.9038
	CSF	0.9070	0.9231	0.8985	0.9125	0.9254	0.9443
	Total	0.8952	0.8980	0.9365	0.9447	0.9398	0.9519
7%	WM	0.7912	0.7461	0.8554	0.9066	0.8675	0.9276
	GM	0.7909	0.7176	0.8595	0.8629	0.8437	0.8900
	CSF	0.8770	0.8971	0.8979	0.8954	0.8893	0.9228
	Total	0.8764	0.8719	0.9133	0.9290	0.9298	0.9354

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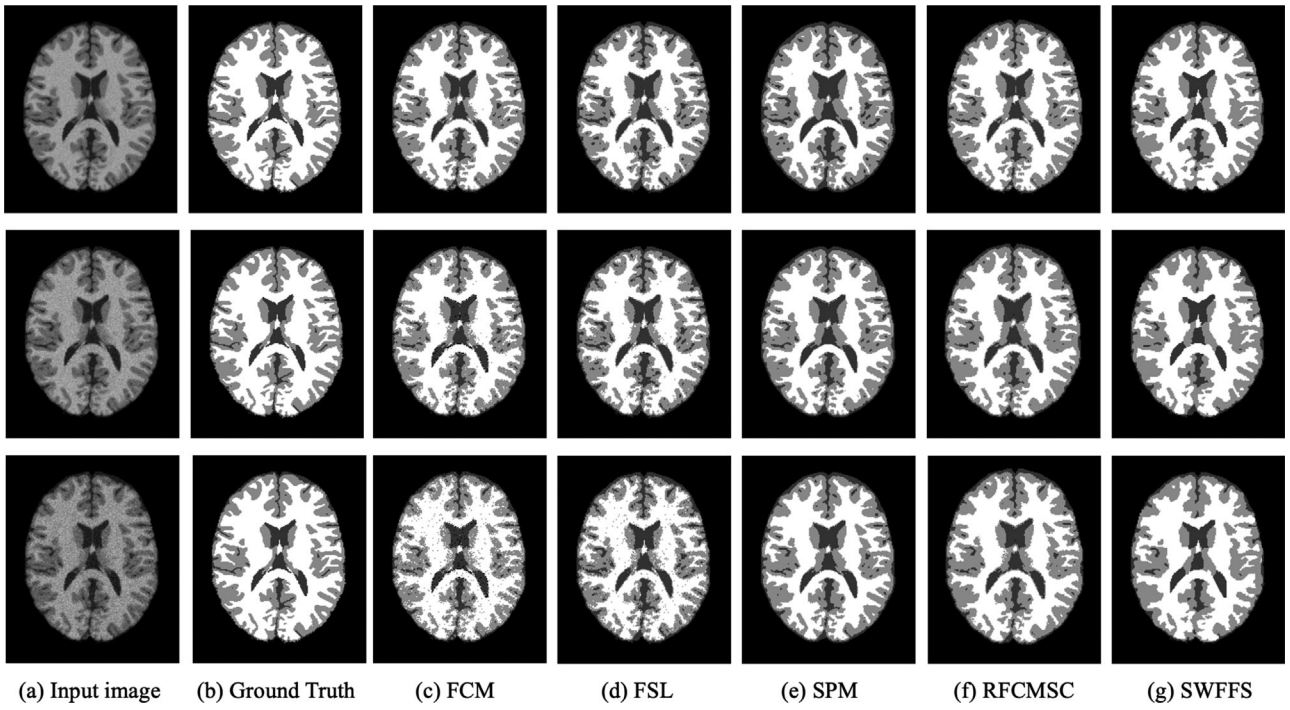


FIGURE 5 The segmentation results of brain MR images from BrainWeb dataset with noise levels of 3%, 5% and 7% from up to bottom

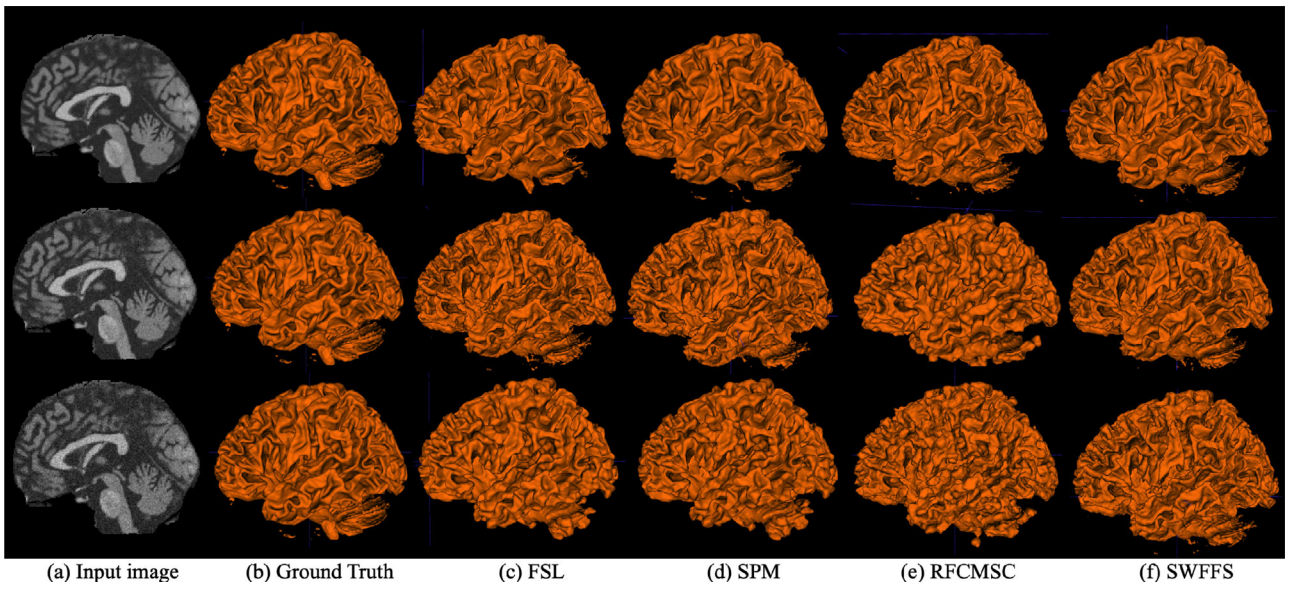


FIGURE 6 The rendering brain surfaces of segmentation results (from sagittal view). From top to bottom, the noise levels are 3%, 5% and 7%, respectively

obtain higher total SAs and tissue SAs than FCM, FSL, SPM, AWSFCM and RFCMSC, indicating that it performs better than the other algorithms.

In Table 3, the SAs of CSF of all algorithms are low because the ground truth does not include the CSF in GM. Compared with FCM, FSL, SPM, AWSFCM and RFCMSC, SWFFS can obtain higher total SAs and WM/GM SAs, suggesting that the segmentation results are closer to the ground truth. In Tables 2 and 3, the total SA values are higher than the average of all tissue SA values because the total SAs computed on the brain MR images includes parts of the background.

We further adopt a popular measurement, dice similarity coefficient (DSC), to quantitatively assess the performance of these segmentation algorithms. For each tissue, DSC computes the ratio between segmentation results and ground truth as fol-

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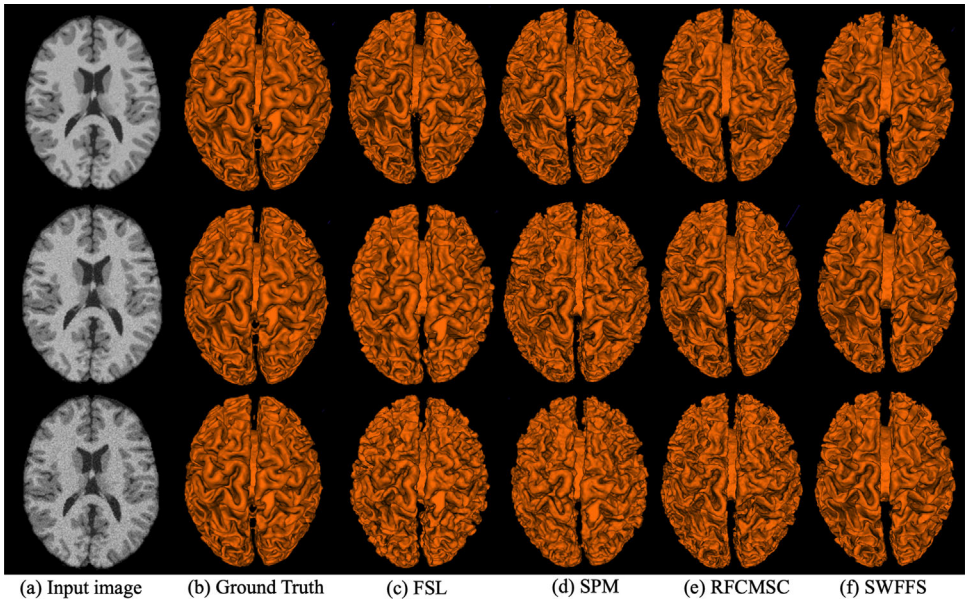


FIGURE 7 The rendering brain surfaces of segmentation results (from axial view). From top to bottom, the noise levels are 3%, 5% and 7%, respectively

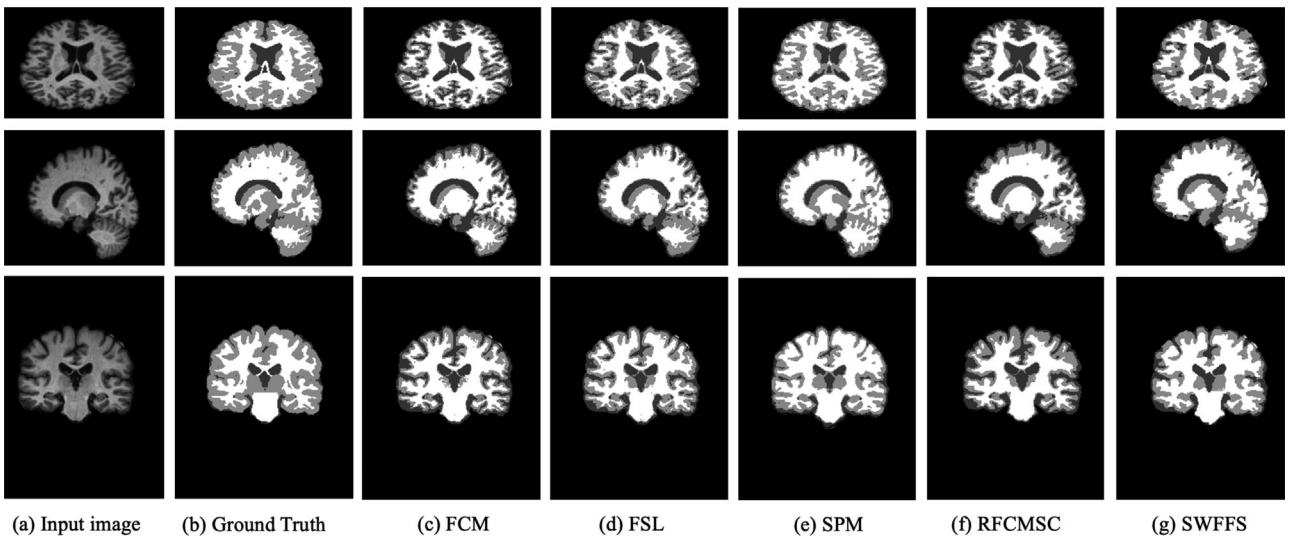


FIGURE 8 The segmentation results of brain MR image from IBSR-18 dataset (Volume: No. 8). The views are axial, sagittal and coronal from up to bottom

lows:

$$DSC = \frac{2 \times TP}{2 \times TP + FP + FN}, \tag{12}$$

FP, FN and TP are the numbers of false-positive, false-negative and true-positive voxels, respectively. The values of DSC can range from 0 to 1, and high values mean more similarity to the ground truth. The DSC values of the four algorithms performed on BrainWeb and IBSR datasets are shown in Tables 4 and 5, respectively. In Table 4, SWFFS can obtain higher DSCs than FCM, FSL, SPM, AWSFCM and RFCMSC, indicating that SWFFS can achieve more accurate tissue structures. As shown in Table 5, the DSCs of CSF of all algorithms are low because the ground truth considers the CSF in GM as GM. SWFFS can obtain higher total DSCs and WM/GM DSCs than the other algorithms, indicating that the segmentation results are closer to the ground truth.

4 | CONCLUSION

We design a weighted feature fusion clustering segmentation method. First, we pre-process brain MR images by generating 3D SLIC supervoxels. Subsequently, we select features of each supervoxel on the basis of appearance, shape, and space location, to capture general and detailed information of each supervoxel. We propose a weighted similarity network fusion method to comprehensively capture the common and

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TABLE 3 Comparative SA analysis of different algorithms on IBSR-18

Noise	Tissue	FCM	FSL	SPM	AWSFCM	RFCMSC	SWFFS
IBSR	WM	0.9055	0.7804	0.9054	0.8994	0.9032	0.9106
-18	GM	0.7805	0.8647	0.8062	0.8638	0.8297	0.8773
(No.	CSF	0.6161	0.6821	0.6963	0.6910	0.6701	0.6808
1)	Total	0.8673	0.8620	0.8724	0.8651	0.8759	0.8820
IBSR	WM	0.8538	0.8339	0.9047	0.9004	0.9025	0.9137
-18	GM	0.7541	0.8197	0.8073	0.8579	0.8403	0.8601
(No.	CSF	0.6323	0.7102	0.7762	0.7250	0.6734	0.7482
8)	Total	0.8321	0.8469	0.8755	0.8799	0.8810	0.8935

TABLE 4 Comparative DSC analysis of different algorithms on brainweb

Noise	Tissue	FCM	FSL	SPM	AWSFCM	RFCMSC	SWFFS
	WM	0.8864	0.8593	0.8811	0.8927	0.8892	0.9059
3%	GM	0.8776	0.7602	0.8577	0.8864	0.8761	0.8900
	CSF	0.7639	0.7837	0.8684	0.8975	0.8703	0.9141
	WM	0.8431	0.7893	0.8615	0.8926	0.8790	0.9015
5%	GM	0.8217	0.7446	0.8534	0.8501	0.8648	0.8656
	CSF	0.7566	0.7085	0.8542	0.8898	0.8632	0.9079
	WM	0.8655	0.8369	0.7990	0.8762	0.8704	0.8905
7%	GM	0.7913	0.7424	0.8330	0.8333	0.8386	0.8417
	CSF	0.7321	0.6568	0.7718	0.8631	0.8421	0.8969

TABLE 5 Comparative DSC analysis of different algorithms on IBSR-18

Noise	Tissue	FCM	FSL	SPM	AWSFCM	RFCMSC	SWFFS
IBSR	WM	0.7788	0.7312	0.7776	0.8465	0.8317	0.8599
-18	GM	0.7366	0.7286	0.7260	0.7793	0.7625	0.7880
(No. 1)	CSF	0.6081	0.6294	0.6677	0.6710	0.6334	0.6998
IBSR	WM	0.8054	0.8302	0.8541	0.8793	0.8758	0.8982
-18	GM	0.6872	0.7744	0.7376	0.7824	0.7853	0.8024
(No. 8)	CSF	0.5826	0.6021	0.6868	0.6955	0.6720	0.6998

complementary characteristics of the similarities of supervoxels. This method can automatically adjust the weight of shape feature in each supervoxel by introducing the supervoxel-level intensity inhomogeneous factor. Finally, we perform the spectral clustering on the fused similarity matrix, and use a diffusion post-processing method to smoothen the segmentation boundaries. Experimental results obtained by applying our techniques to various brain MR images demonstrate that the SWFFS method can improve the accuracy and robustness of the brain MR image segmentation.

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CONFLICT OF INTEREST

The authors have declared no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in BrainWeb at http://www.brainweb.com and https://www.nitrc.org/projects/ibsr.

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REFERENCES

1. Roy, S., Maji, P.: Rough segmentation of coherent local intensity for bias induced 3-D MR brain images. Pattern Recognit. 97, 106997 (2020)
2. Fan, J., Cao, X., Yap, P.T., Shen, D.: Birnet: Brain image registration using dual-supervised fully convolutional networks. Med. Image Anal. 54, 193–206 (2019)
3. Xia, J., Wang, F., Wu, Z., Wang, L., Zhang, C., Shen, D., et al.: Mapping hemispheric asymmetries of the macaque cerebral cortex during early brain development. Hum. Brain Mapp. 41(1), 95–106 (2020)
4. Halder, A., Talukdar, N.A.: Robust brain magnetic resonance image segmentation using modified rough-fuzzy c-means with spatial constraints. Appl. Soft Comput. 85, 105758 (2019)
5. He, K., Cao, X., Shi, Y., Nie, D., Gao, Y., Shen, D.: Pelvic organ segmentation using distinctive curve guided fully convolutional networks. IEEE Trans. Med. Imaging 38(2), 585–595 (2018)
6. Erdil, E., Ghani, M.U., Rada, L., Argunsah, A.O., Unay, D., Tasdizen, T., et al.: Nonparametric joint shape and feature priors for image segmentation. IEEE Trans. Image Process. 26(11), 5312–5323 (2017)
7. Moyà Alcover, G., Elgammal, A., Jaume-i Capó, A., Varona, J.: Modeling depth for nonparametric foreground segmentation using RGBD devices. Pattern Recognit. Lett. 96, 76–85 (2017)
8. Huo, J., Wu, J., Cao, J., Wang, G.: Supervoxel based method for multi-atlas segmentation of brain MR images. NeuroImage 175, 201–214 (2018)
9. Prakash, R.M., Kumari, R.S.S.: Spatial fuzzy c means and expectation maximization algorithms with bias correction for segmentation of MR brain images. J. Med. Syst. 41(1), 1–9 (2017)
10. Ji, Z., Liu, J., Cao, G., Sun, Q., Chen, Q.: Robust spatially constrained fuzzy c-means algorithm for brain mr image segmentation. Pattern Recognit. 47(7), 2454–2466 (2014)
11. Chen, Y., Zhang, H., Zheng, Y., Jeon, B., Wu, Q.J.: An improved anisotropic hierarchical fuzzy c-means method based on multivariate student t-distribution for brain mri segmentation. Pattern Recognit. 60, 778–792 (2016)
12. Fan, J., Yang, J., Ai, D., Xia, L., Zhao, Y., Gao, X., et al.: Convex hull indexed Gaussian mixture model (ch-gmm) for 3D point set registration. Pattern Recognit. 59, 126–141 (2016)
13. Wang, L., Nie, D., Li, G., Puybareau, É., Dolz, J., Zhang, Q., et al.: Benchmark on automatic six-month-old infant brain segmentation algorithms: The iseg-2017 challenge. IEEE Trans. Med. Imaging 38(9), 2219–2230 (2019)
14. Kong, Y., Wu, J., Yang, G., Zuo, Y., Chen, Y., Shu, H., et al.: Iterative spatial fuzzy clustering for 3D brain magnetic resonance image supervoxel segmentation. J. Neurosci. Methods 311, 17–27 (2019)
15. Wang, M., Liu, X., Gao, Y., Ma, X., Soomro, N.Q.: Superpixel segmentation: A benchmark. Signal Process. Image Commun. 56, 28–39 (2017)
16. Xu, C., Corso, J.J.: Libsvx: A supervoxel library and benchmark for early video processing. Int. J. Comput. Vision 119(3), 272–290 (2016)
17. Tang, D., Fu, H., Cao, X.: Topology preserved regular superpixel. In: 2012 IEEE International Conference on Multimedia and Expo, pp. 765–768. IEEE, Piscataway, NJ (2012)
18. Shen, J., Du, Y., Wang, W., Li, X.: Lazy random walks for superpixel segmentation. IEEE Trans. Image Process. 23(4), 1451–1462 (2014)
19. Neubert, P., Protzel, P.: Compact watershed and preemptive slic: On improving trade-offs of superpixel segmentation algorithms. 2014 22nd

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International Conference on Pattern Recognition, pp. 996–1001. IEEE, Piscataway, NJ (2014)
20. Conrad, C., Mertz, M., Mester, R.: Contour-relaxed superpixels. International Workshop on Energy Minimization Methods in Computer Vision and Pattern Recognition, pp. 280–293. Springer, Berlin (2013)
21. Papon, J., Abramov, A., Schoeler, M., Worgotter, F.: Voxel cloud connectivity segmentation-supervoxels for point clouds. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2027–2034. IEEE, Piscataway, NJ (2013)
22. Liu, Y.J., Yu, C.C., Yu, M.J., He, Y.: Manifold slic: A fast method to compute content-sensitive superpixels. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 651–659. IEEE, Piscataway, NJ (2016)
23. Mahata, N., Kahali, S., Adhikari, S.K., Sing, J.K.: Local contextual information and Gaussian function induced fuzzy clustering algorithm for brain mr image segmentation and intensity inhomogeneity estimation. Appl. Soft Comput. 68, 586–596 (2018)
24. Dera, D., Bouaynaya, N., Fathallah Shaykh, H.M.: Automated robust image segmentation: Level set method using nonnegative matrix factorization with application to brain MRI. Bull. Math. Biol. 78(7), 1450–1476 (2016)
25. Kong, Y., Deng, Y., Dai, Q.: Discriminative clustering and feature selection for brain MRI segmentation. IEEE Signal Process Lett. 22, 573–577 (2015)
26. Scovanner, P., Ali, S., Shah, M.: A 3-dimensional sift descriptor and its application to action recognition. Proceedings of the 15th ACM International Conference on Multimedia, pp. 357–360. (2007)
27. Zhu, Z., Zheng, M., Qi, G., Wang, D., Xiang, Y.: A phase congruency and local Laplacian energy based multi-modality medical image fusion method in NSCT domain. IEEE Access 7, 20811–20824 (2019)
28. Wang, B., Mezlini, A.M., Demir, F., Fiume, M., Tu, Z., Brudno, M., et al.: Similarity network fusion for aggregating data types on a genomic scale. Nat. Methods 11(3), 333 (2014)
29. Kwan, R.S., Evans, A.C., Pike, G.B.: MRI simulation-based evaluation of image-processing and classification methods. IEEE Trans. Med. Imaging 18(11), 1085–1097 (1999)
30. Xia, J., Wang, F., Meng, Y., Wu, Z., Wang, L., Lin, W., et al.: A computational method for longitudinal mapping of orientation-specific expansion of cortical surface in infants. Med. Image Anal. 49, 46–59 (2018)
31. Bezdek, J.C.: Pattern Recognition with Fuzzy Objective Function Algorithms. Springer Science & Business Media, Berlin (2013)
32. Jenkinson, M., Beckmann, C.F., Behrens, T.E., Woolrich, M.W., Smith, S.M.: FSL. Neuroimage 62(2), 782–790 (2012)
33. Ashburner, J., Friston, K.J.: Computing average shaped tissue probability templates. Neuroimage 45(2), 333–341 (2009)
34. Mishro, P., Agrawal, S., Panda, R., Abraham, A.: A novel type-2 fuzzy c-means clustering for brain MR image segmentation. IEEE Trans. Cybern. 51, 3901–3912 (2020)

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